that simple to replicate with an algorithm: if we want to avoid resorting to computer vision (and we usually do, for this kind of tasks[9]) we need to restrict the way we draw edges, for instance limiting to straight-line segments or Bézier curves, so that we can use math formulas to compute if they intersect. Still, the computational effort needed to check the number of intersections on a large graph remains considerable.

And this is just for a single embedding: determining if a graph is non-planar means proving that for any possible embedding we can come up with, there is at least an intersection.

We have already introduced Kuratowski's work on planar graphs, providing the first method to determine if a graph is planar.

Planarity, however, had been already studied for a long time, and in fact Euler, in 18th century, came up with an invariant (proved only in 1811 by Cauchy) providing necessary conditions for a graph to be planar.

Although these conditions are not sufficient, so they can't be used to prove planarity, they are cheap to compute and when violated they rule out planarity.

The two conditions we can more easily implement in our tests are:

- Given a simple, connected graph $G=(V,E)$ with at least 3 vertices, G is planar only if $|E| \leq 3|V| - 6$;